INCH-POUND

MIL-P-63225A (AR)
AMENDMENT 3
12 JULY 1989
SUPERSEDING
AMENDMENT 2
18 May 1987

MILITARY SPECIFICATION

PROJECTILE, GUIDED, 155MM, M712
CONTROL SECTION FOR

This amendment forms a part of Military Specification MIL-P-63225A (AR), dated 26 April 1984, and is approved for use within the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.1: Add the following Military Specifications:

"MIL-STD-6866 - Inspection, Liquid Penetrant
MIL-H-6875 - Heat Treatment of Steels (Aerospace Practice, Process for)
MIL-STD-2154 - Inspection, Ultrasonics, Wrought Metals, Process for
MIL-I-25135 - Inspection Materials, Penetrant
MIL-STD-45662 - Calibration Systems Requirement"

Add the following paragraph:

"2.2 Other publications. The following documents form a part of this specification to the extent specified herein, unless otherwise indicated, the issue in effect on the date of invitation for bids shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E4 - Verification of Testing Machines
E8 - Tension Testing of Metallic Materials
E18 - Tests for Rockwell Hardness of Metallic Materials
E139 - Recommended Practice for Conducting Creep, Creep-Rupture and Stress Rupture Tests of Metallic Materials
A370 - Standard Method and Definitions for Mechanical Testing of Steel Products
E399 - Plane Strain Fracture Toughness of Metallic Materials

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(Copies may be purchased directly from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

ANSI-B46.1 - Surface Texture

(Application for copies should be addressed to the American Material Standards Institute, 1430 Broadway, New York, New York, 10018).

PAGE 3

3.3.1.1.4 Delete in its entirety and substitute the following:

"3.3.1.1.4 Pneumatic power requirements. The control section shall require helium gas per BB-H-1168 or nitrogen gas per BB-N-411 at 800 psi minimum for non-flight testing."

PAGE 4

3.3.1.3.1 Insert the following after the first sentence: "Adjustment of the fin lock is permissible to meet these requirements."

PAGE 6

Delete Figure 1 in its entirety and substitute the included Figure 1.

PAGE 9

Add the following new paragraphs:

"3.3.3 Welding of control housing. The weld zone shall meet the strength requirements of the finished item.

3.3.4 Control housing integrity.

3.3.4.1 Ultrasonic examination. The weld zone shall be ultrasonically inspected using an angle beam method. No discontinuity shall be acceptable which is more than 1DB below that obtained from the least responsive notch in the reference standard per Figure 6.

3.3.4.2 Fluorescent penetrant examinations. There shall be no indications of cracks, laps, seams, or tears when fully machined part is inspected by fluorescent penetrant inspection.

3.3.5 Mechanical properties. Mechanical properties of the Control Housing shall be as required on the applicable drawing.

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Nominal Rated Values (Vdc)

	Y/R	Pitch
Input Command Volts	10.0	7.50
Feedback Volts	11.10	8.32

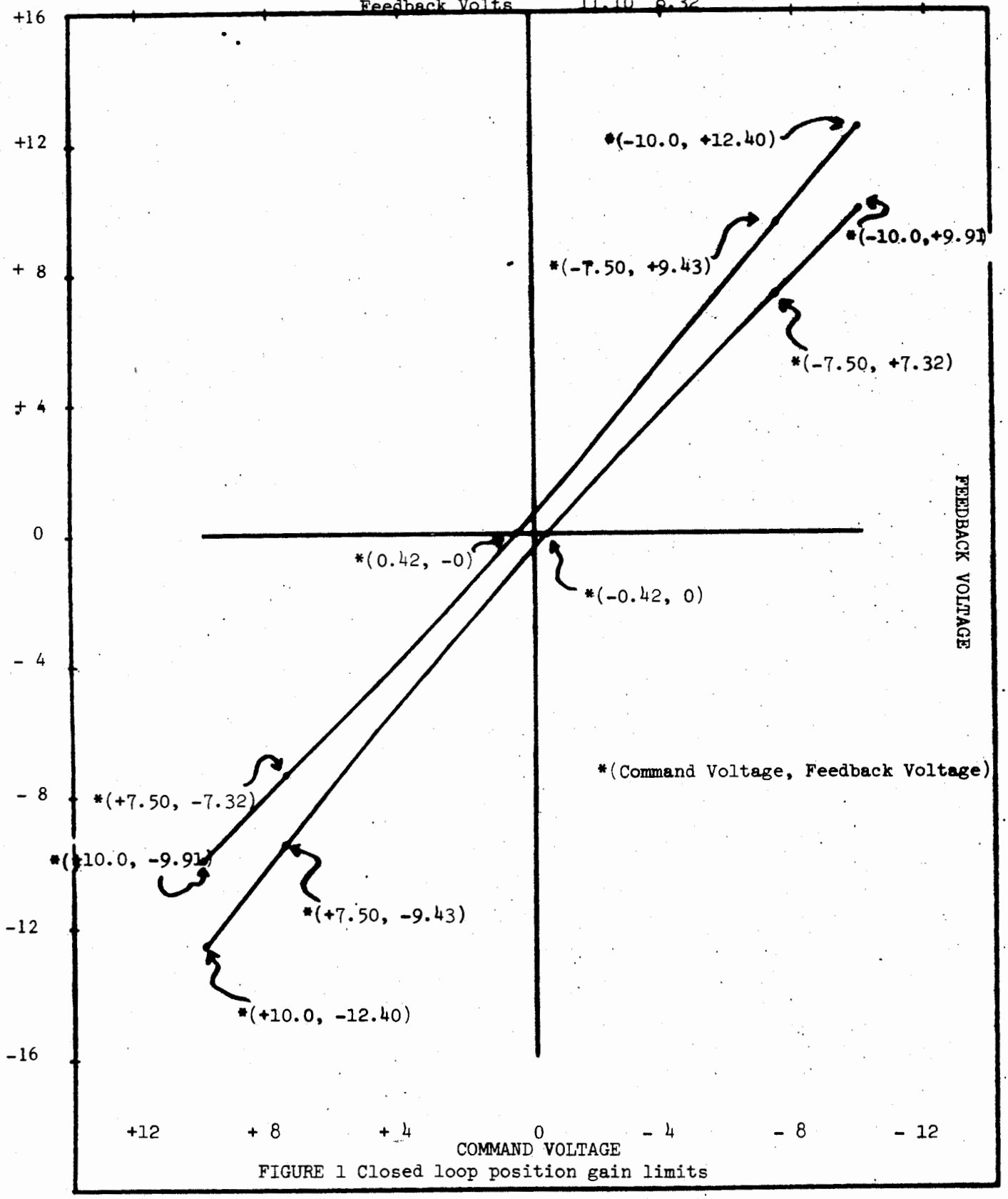


FIGURE 1 Closed loop position gain limits

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3.3.5.1 Fracture toughness. When material exceeds the ultimate strength requirement of 280,000 psi but is less than 290,000 psi, it must have a fracture toughness of at least 40 KSI in $^{1}/_{2}$."

PAGE 14

Add the following paragraphs:

"4.4.1.1 Weld lot formation. Control housing weld lots shall not exceed one mill heat of steel, 400 weldments or the production yield of ten (10) consecutive working days, whichever is smallest. Additionally weld lots shall be restricted to components welded and stress relieved in the same equipment, same process, with the same pressure, the same RPM and with the same flywheel mass. All sections of the housing shall be welded from the same heat of steel. The contractor may submit an alternate procedure for weld lot formation and inspection of weld lots, to the procuring activity for approval.

4.4.1.2 Heat treatment lot formations. Heat treatment lots shall be defined as control housings heat treated in one continuous run (see 4.4.1.2.1) and shall not exceed one mill heat of steel, or 300 components or 24 hours production whichever is smallest. Additionally, heat treatment lots shall be restricted to components heat treated in the same heat treating equipment, at the same temperature, for the same length of time and in one unchanged process. When a batch type furnace is used, the heat treatment inspection lot shall be further restricted to those components heat treated at the same time in one batch. Batches may be combined into large heat treatment lots after qualifying with individual batches, provided qualifying procedures are submitted to and approved by the Government and provided all the other restrictions detailed above are met. Test samples shall be selected from the heat treatment inspection lot after final heat treatment. Components in heat treatment lots shall remain grouped together and identified by lot until mechanical property tests have been completed. Components from heat treatment inspection lots which comply with applicable requirements may be subsequently regrouped into final assembly lots. The contractor may submit an alternate procedure for heat treatment lot formation and inspection of the heat treatment lot to the procuring activity for approval.

4.4.1.2.1 Continuous run. A continuous run is defined to be a run which involves only one part (i.e. control housing, warhead body, or electronic housing), with no interruption of the scheduled flow of parts into or out of the system, and no changes in the parameters of any furnace or bath.

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PAGES 27 and 28

Delete page 27 and 28 and replace with the following pages:

PAGE 38

Paragraph 4.4.2.18, Change Drawing Number from "9308076" to "9308129".

PAGE 57

* 4.5.3.6.1 Add the following sentence to the end of the paragraph: "Adjustment of the fin lock is permissible to meet these requirements."

PAGE 61

Add the following paragraphs:

"4.5.8 Mechanical properties of control housing.

4.5.8.1 Hardness. Each control housing shall be Rockwell hardness tested per paragraph 4.5.10 and this value used for selection of tension test samples of 4.5.8.2.

4.5.8.2 Tensile test. Two control housings from each heat treatment lot shall be selected for this test (one body with the highest hardness reading and one with the lowest hardness reading). From each of these bodies specimens shall be tested per paragraph 4.5.11 to determine compliance with ultimate tensile strength, yield strength and elongation requirements. Failure of any tensile specimens shall be cause for rejection of the heat treat lot.

4.5.8.2.1 Retest. If the high hardness body fails the yield strength or minimum ultimate strength requirement, the entire heat treatment lot shall be rejected. In the event the lowest hardness body fails to meet the yield strength or minimum ultimate strength requirement, higher hardness values may be selected and tested in accordance with 4.5.8.2 in order to establish the acceptable minimum hardness. Once this minimum hardness has been established, all bodies with hardness value below the new minimum shall be rejected. If the low hardness body fails the elongation requirement, the entire heat treatment inspection lot shall be rejected. In the event the highest hardness body fails to meet the elongation requirement, lower hardness values may be selected and tested in accordance with 4.5.8.2 in order to establish the acceptable maximum hardness. Once this maximum hardness has been established, all bodies with hardness values above the new maximum shall be rejected.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER 9308050
4.4.2.9	Control housing				NEXT HIGHER ASSEMBLY 9308000
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Penetrant inspection for cracks		100%	3.3.4.2	4.5.13
2	Minimum wall thickness center section of housing		100%	Dwg	Gage
3	Minimum thickness between keyway and crush pin hole		100%	Dwg	Gage
Major					
101	Control housing material		N/A	Dwg	Vendor lot certifications
102	Hardness		100%	Dwg	4.5.10
103	Mechanical properties		4.5.8.2	3.3.6	4.5.11
104	Fracture toughness (see note 1)		4.5.15.1	3.3.5.1	4.5.15.2
105	Mechanical properties of weld		4.5.9	3.3.3	4.5.12
106	Ultrasonic inspection of weld zone		2.5%	3.3.4.1	4.5.14
107	Width and depth of snap ring groove		0.40%	Dwg	Gage
108	Angle of forward face of snap ring groove		0.40%	Dwg	Gage
109	Maximum wall thickness center section of housing		100%	Dwg	Gage
110	Outside diameter		100%	Dwg	Gage
111	Two bourrelet diameters		0.40%	Dwg	Gage
112	Keyway dimensions		100%	Dwg	Gage

NOTES: 1. This test is only required when the material tested exceeds the ultimate strength requirement of 280,000 psi but is less than 290,000 psi.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 2 OF 3	DRAWING NUMBER
4.4.2.9	Control housing		9308050
			NEXT HIGHER ASSEMBLY 9308000

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Depth of SES shoulder		0.65%	Dwg	Gage
202	Distance from aft end to forward shoulder		0.65%	Dwg	Gage
203	Distance from aft end to aft shoulder		0.65%	Dwg	Gage
204	Run out of aft closure interface shoulder with respect to the axis of the forward and aft I.D.s		0.65%	Dwg	Gage

NOTES:

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4.5.8.3 Elongation. Acceptance shall be based on the average elongation of all tensile specimens selected from each housing. If a tensile specimen fails within the weld zone, then that particular elongation value shall not be used in computing the average elongation. If the average elongation is less than 10 percent, then the lot represented by this test sample shall be rejected.

4.5.9 Mechanical properties of control housing weld.

4.5.9.1 Tensile testing. A housing that contains the first weldment from each weld lot and a different housing that contains the last weldment shall be selected for this test. Four (4) tensile bars shall be selected from each weld of each housing per paragraph 4.5.12 to determine compliance with tensile strength, yield strength and elongation requirements. Failure to meet the requirements of 3.3.3 shall be cause for rejection.

4.5.10 Hardness. In order to determine uniformity of heat treatment, each body shall be subjected to a hardness test after final heat treatment with a Rockwell Hardness Tester or any other method approved by the procuring activity. Readings shall be taken on the rear face of the control housing. The test shall be made using the procedure for performing this test as prescribed by ASTM E-18 and ASTM A-370.

4.5.11 Tensile strength, yield strength, and percent elongation. The hardest and softest bodies as determined by the hardness test in 4.5.10 from each heat treat lot shall be subjected to this test. The criteria for performing the test shall be as prescribed by ASTM E-8.

4.5.11.1 Control housing. Four (4) tensile specimens shall be selected from each housing. The specimens shall be selected approximately 90° apart and shall be the largest obtainable longitudinal specimenas prescribed by ASTM E-8.

4.5.12 Tensile testing of weld. Four (4) tensile specimens shall be selected from the weld section, 90° apart, per 4.5.9.1. The specimens shall be the largest obtainable round type specimen as prescribed by ASTM E-8. The weld cross-section shall be in the middle half of the gage length of the tensile specimen. The samples shall be taken after welding and stress relieving. The samples shall then be heat treated to the requirements of the control housing and tested per ASTM E-8. Failure to meet the applicable strength and elongation requirements shall be cause for rejection of the lot.

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4.5.13 Fluorescent penetrant inspection. Fluorescent pentrant inspection shall be performed on 100 percent of the housings after the final machining and heat treating. Inspection shall be in accordance with Group V, MIL-I-25135 and MIL-STD-6866. Both sides of the part shall be examined. Parts examined by penetrant and found acceptable shall be ink stamped or stenciled with the letter "P". The marking shall withstand normal handling.

4.5.14 Ultrasonic inspection. Each lot of housings shall be ultrasonically examined after weld flash removal and stress relief.

4.5.14.1 Procedure. Ultrasonic testing shall be conducted following the general principles outlined in MIL-STD-2154. Calibration of the ultrasonic system using reference standards containing simulated discontinuities shall be performed at the beginning of each test run of given part, after each 30 minutes during the run, at the end of each run, and any time that a malfunction is suspected. If any check reveals that the system is not functioning properly, all of the material that has been examined since the last acceptable check shall be reexamined. Calibration shall be performed by obtaining response from each of the simulated discontinuities associated with the zone under test, setting the various controls such that the least of the responses triggers the alarm, and then increasing the system gain by 1 DB for production part testing.

4.5.14.2 Ultrasonic inspection standards. Reference standards shall be of the same thickness, material and configuration as the parts being examined. Simulated discontinuities will be circular segment notches circumferentially cut into the I.D. and O.D. of each weld zone. See Figure 6. For each weld zone I.D. and O.D. there will be 3 notches. One at the near side of the weld zone, one at their nominal center of the weld zone, and one in the far side of the weld zone. The reference standard shall be appropriately treated to inhibit corrosion."

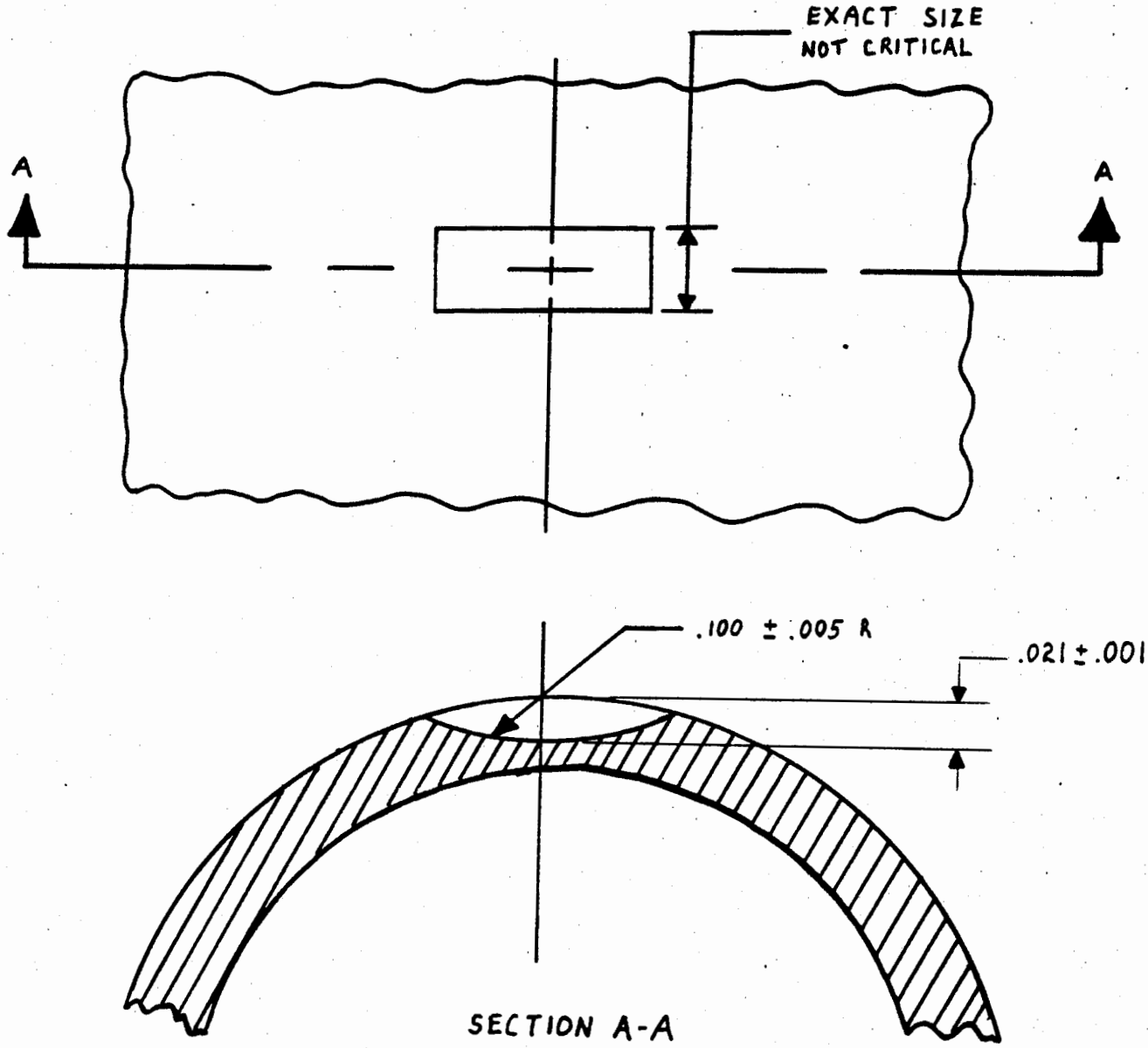
4.5.15 Fracture toughness testing.

4.5.15.1 Slow bend fracture toughness test. When a specimen taken from a housing as selected per paragraph 4.5.8.2 is tested and exceeds the ultimate strength requirement of 280,000 psi but is less than 290,000 psi, slow bend (three point bending) fracture toughness tests are required. The bend testing shall be conducted in accordance with the requirements and procedures of ASTM E399. The specimans shall be tested with acceptance based on the average fracture toughness. If the average fracture toughness is below the minimum allowable, the heat treated lot represented shall be rejected or subjected to retemper and retest for all mechanical properties.

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4.5.15.2 Fracture toughness specimen selection. Three(3) specimens shall be taken from the housing that exhibited the high ultimate strength readings (above 280 Kpsi). If both housings exhibit high ultimate strength readings then the housing with the highest hardness reading shall be used for specimen selection. The three specimens shall be selected in the longitudinal direction and taken from the aft end of the housing. The specimen shall have a crack plane orientation L - C as shown in Figure 7. The specimen dimensions shall be within the tolerances shown in Figure 9.

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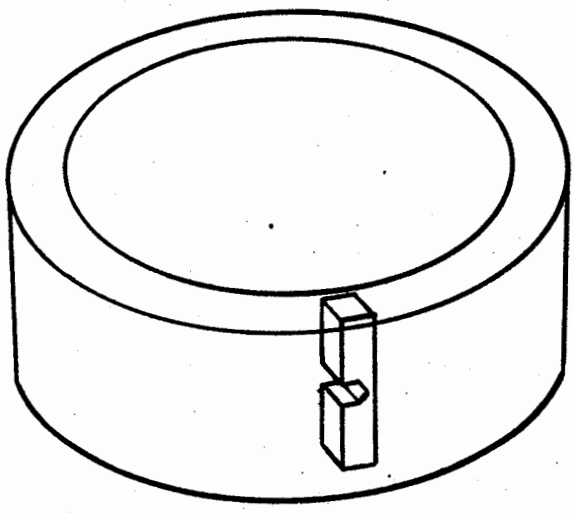


NOTES:

1. NOTCH CROSS SECTIONAL AREA 0.00174 SQ. IN.
2. I.D. NOTCH HAS SAME DIMENSIONS AS SHOWN ABOVE

FIGURE 6

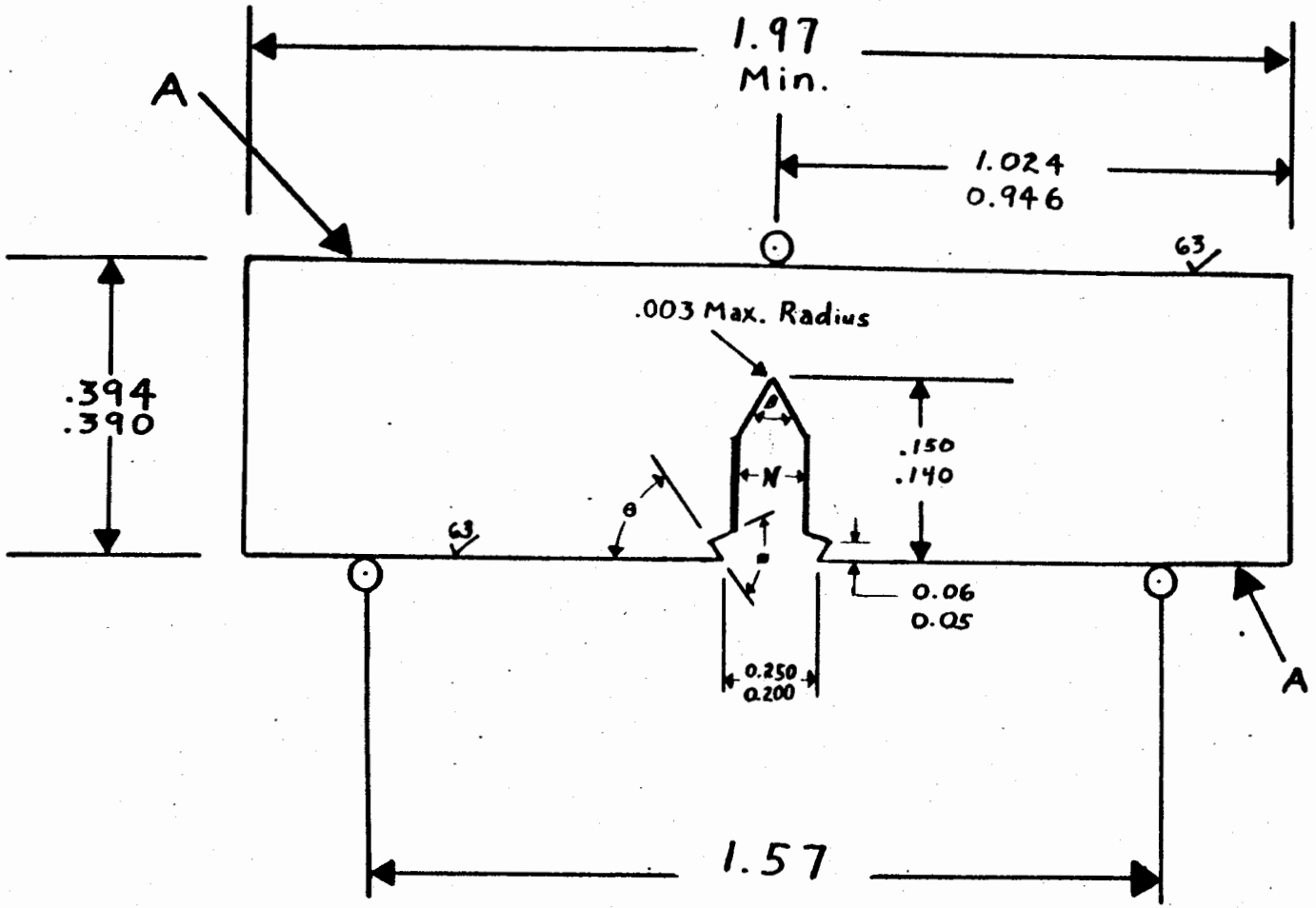
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CRACK PLANE ORIENTATION FOR COPPERHEAD CONTROL SECTION

FIGURE 7

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$45° \leq \beta \leq 60°$

$45° \leq \theta \leq 60°$

$\theta \leq \phi \leq 90°$

$N = .070$ max.

NOTE 1-ALL DIMENSIONS IN INCHES
NOTE 2-"A" SURFACES SHALL BE PERPENDICULAR OR PARALLEL AS APPLICABLE TO WITHIN 0.001 IN TIR.
NOTE 3-INTEGRAL OR ATTACHABLE KNIFE EDGES FOR CLIP GAGE ATTACHMENT MAY BE USED
NOTE 4-CRACK STARTER NOTCH SHALL BE PERPENDICULAR TO SPECIMEN SURFACES TO WITHIN $\pm 2^{0}$.

FIGURE 8

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"6.7 Weld upset (Advisory). It has been found that acceptable welds have been produced with upset lengths of between 0.230 to 0.305 inches. Mating surfaces of parts to be inertia welded should be of the same inside diameter and same wall thickness.

6.8 Holding grip marks (Advisory). It is recommended that holding grips be so configured as to not allow slippage of component more than 0.100 inch in the welding operation. Furthermore, adequate surface should be provided to avoid the slippage, and to resist the torque developed during welding, without deforming the part."

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Preparing activity
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